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The use of aggregation pheromone to enhance dissemination of *Beauveria bassiana* for the control of the banana weevil in Uganda

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Abstract

Candidate strains of *Beauveria bassiana* were identified for use in integrated pest management of the banana weevil *Cosmopolites sordidus*. Horizontal field transmission of *B. bassiana* between banana weevils using different delivery systems, including aggregation pheromones, was investigated. We observed that infected weevils could transmit the fungal pathogen to healthy individuals. Most dead weevils (52%) due to *B. bassiana* infection were found at the base of banana plants in the leaf sheath or in the soil near banana plants. Significantly more weevils died from the pathogen in plots where *B. bassiana* was applied in combination with the pheromone than where it was applied alone. Our data demonstrate that *C. sordidus* aggregation pheromone can be a valuable tool to enhance the dissemination of *B. bassiana* for the control of *C. sordidus*.

Keywords: Aggregation pheromone, banana, *Beauveria bassiana*, *Cosmopolites sordidus*, dispersal, IPM

Introduction

The banana weevil *Cosmopolites sordidus* (Germar) (Coleoptera: Curculionidae) is a principal pest of highland bananas and plantains in Africa (Gold et al. 2001). The adults are free-living and insert their eggs into the base of the banana plant (Abera et al. 2000). The larvae bore in the rhizome (corm), reducing nutrient uptake and weakening plant anchorage. *Cosmopolites sordidus* attack causes premature plant death, snapping, toppling, reduced bunch weights, plant disappearance and shortened stand life (Gold et al. 2001). Yield losses of 100% have been observed in farmers' fields (Sengooba 1986), while losses of more than 40% have been reported in on-station

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trials (Rukazambuga et al. 1998; Gold et al. 2004a). The banana weevil has been a major factor in the decline of highland banana in central Uganda (Gold et al. 1999).

Cosmopolites sordidus is a difficult pest to control. The immature stages are within the plant, often below the soil level, making them largely inaccessible to natural enemies (Gold et al. 2001). Therefore, control strategies often target adults. *Cosmopolites sordidus* adults are hydrotropic, most often found near banana mats (a group of banana plants arising from a common rhizome) or crop residues, nocturnally active, relatively sedentary and they rarely fly (Gold et al. 2001, 2004b). Few disperse more than 50 m in 3 months (Gold et al. 2001, 2004b). They are seldom observed outside banana stands.

Microbial control offers a plausible method for managing cryptic pests such as the banana weevil. Strains of the entomopathogen *Beauveria bassiana* (Balsamo) Vuillemin (Ascomycota: Hypocreales) (e.g., Ugandan isolate G41) with efficacy against banana weevil adults have been identified in most important banana growing regions (Gold et al. 2001). Mortalities of up to 100% due to *B. bassiana* (3×10^9 conidia/g) infection have been reported in the laboratory following topical application (Nankinga 1994). Infected weevils usually show signs of mycosis within 2 weeks. Only limited work, however, has been done on the control potential of *B. bassiana* against banana weevil under field conditions.

In Ghana, Godonou et al. (2000) successfully protected newly planted suckers against *C. sordidus* attack by applying *B. bassiana* (formulated with oil palm residues) to planting holes and around young banana mats. These authors did not investigate sustained protection over the entire crop cycle. In Uganda, multiple applications of maize-based formulations of *B. bassiana* at the base of banana stands reduced *C. sordidus* populations and damage to harvested plants (Nankinga 1999). However, the use of a food crop for a substrate (locally available alternative substrates were not effective) and the amount of *B. bassiana* required for applications at the base of each banana mat were unlikely to be adopted by highland banana growers. Therefore, strategies that can aggregate *C. sordidus* at a limited number of *B. bassiana* delivery sites might be able to reduce costs and prove more feasible. Key questions for such a strategy would be the degree to which adults might be aggregated, the likelihood of *C. sordidus* to spread the pathogen from delivery sites and the ability of infected weevils to transmit the fungus to uninfected weevils (horizontal transmission).

Cosmopolites sordidus are attracted to plant volatiles emanating from crop residues (Budenberg et al. 1993a). This attraction has been exploited through the use of split pseudostem traps (pieces of pseudostem placed flat on the soil) for monitoring and managing the pest (Mitchell 1978; Gold et al. 2002). Application of *B. bassiana* to pseudostem traps in order to aggregate adult *C. sordidus* proved ineffective because the micro-environment under the traps accelerated deterioration of the fungus and loss of viability (Nankinga 1999). Tumuhaise et al. (2004) attempted to aggregate banana weevils using volatiles from ground corm residues, but this method was unsuccessful. By contrast, Treverrow and Bedding (1993) and Treverrow (1994) effectively utilized plant kairomones to aggregate *C. sordidus* for the delivery of entomophagous nematodes by making conical cuts in the corms of banana stumps.

A male aggregation pheromone was demonstrated by Budenberg et al. (1993b). Pheromone-baited traps were observed to aggregate *C. sordidus* on trap mats and adjacent mats in field studies conducted in Uganda (Tinzaara et al. 2005). These authors suggested the use of *C. sordidus* pheromones to aggregate adults at entomopathogen delivery sites. The use of the pheromone as a lure for the

dissemination of *B. bassiana* has been reported to be effective for other beetles (Klein and Lacey 1999; Yasuda 1999; Vega et al. 2000; Dowd and Vega 2003; Kreutz et al. 2004) but not yet for *C. sordidus*. The development of a pheromone trapping system for dissemination of biological control agents would require pheromone-baited traps that allow both entry and exit of *C. sordidus* (Tinzaara et al. 2000). Successful use of pheromone lures to enhance dissemination of *B. bassiana* would also require that large numbers of the pest are attracted to the trap, where they pick up the pathogen and infect healthy individuals outside the trap. This delivery system would have the advantage of reduced costs as only a limited amount of the pathogen is needed in traps and field applications would not be necessary.

Lures containing *C. sordidus* pheromone have been made by Chemtica International (San Jose, Costa Rica) and tested by Alpizar et al. (1999) in Costa Rica and by Tinzaara et al. (2000) in Uganda. Results in Uganda were not as encouraging as those in Costa Rica. One possible difference may have been the relative efficacy of ramp traps (Alpizar et al. 1999) versus pitfall traps (Tinzaara et al. 2000). Thus, it is possible that pheromone lures may have attracted *C. sordidus* to the vicinity of the lures without the weevils actually entering the traps.

The behavior and field distribution of *B. bassiana*-infected *C. sordidus* has not been studied. Thus, there is little information on the degree to which infected weevils can carry the pathogen throughout the insect's habitat and infect conspecifics. Horizontal transmission has been demonstrated in the laboratory (Nankinga 1994; Schoeman and Schoeman 1999; Godonou et al. 2000) but not under field conditions. Detailed information on the effectiveness of the transmission of *B. bassiana* from infected individuals to non-infected conspecifics is necessary to develop a delivery system for the pathogen. As the movement of infected hosts determines the way in which a pathogen is transmitted and spread within a field (Fuxa and Tanada 1987), the behavior of infected *C. sordidus* in the field should also be studied.

The objectives of this study were to: (1) determine the potential for pheromone-baited traps to aggregate *C. sordidus*; (2) determine the effect of *B. bassiana* infection on weevil behavior (dispersal and location) in the field, (3) evaluate the field transmission of *B. bassiana* from infected to uninfected *C. sordidus*, and (4) compare different delivery systems of *B. bassiana* for *C. sordidus* control.

Materials and methods

Study sites

Field studies were conducted in and on the International Institute of Tropical Agriculture's Sendusu Farm (0°32'N, 32°35'E, 1260 metres above sea level (m.a.s.l.)) located 28 km northeast of Kampala, Uganda. Laboratory studies were conducted at the Kawanda Agricultural Research Institute (0°25'N, 32°51'E, 1190 m.a.s.l.) located 12 km north of Kampala. All sites have two rainy seasons (March–May and September–November) with mean annual rainfall of 1200–1300 mm and a mean daily temperature of 21°C.

Source of pheromone and B. bassiana

The synthetic pheromone Cosmolure+ (produced by Chemtica, International, San Jose, Costa Rica) was used in all experiments. It consists of a mixture of four sordidin

((1*S*,3*R*,5*R*,7*S*)-(+) -1-ethyl-3,5,7-trimethyl-2,8-dioxabicyclo[3.2.1]octane) isomers (Mori et al. 1996) that are packaged in poly-ethylene with a release rate of 3 mg/day (A.C. Oehlschlager, pers. comm.). The lures were shipped in tightly sealed plastic bags that were stored in a dark cupboard at 21–25°C. Pheromone packs from opened bags were stored in the freezer at –5°C.

Beauveria bassiana (isolate G41: country of origin is Uganda and the host is soil) (3×10^9 conidia/g) was obtained from the pathology laboratory at KARI in a crushed maize formulation (Nankinga and Moore 2000). This isolate had been previously shown to be effective against *C. sordidus* in both the laboratory and field (Nankinga 1999).

Experiment 1. Dispersal of inoculated and uninoculated C. sordidus under field conditions

This experiment was conducted to determine the influence of *B. bassiana*-infection on *C. sordidus* dispersal within banana fields during the period from June to August 2003. Study plots consisted of two 5-year old banana plots (cv Nabusa, AAA-EA, 42 mats; 3 × 3 m spacing) at Sendusu. The plots were 200 m apart, well weeded and self-mulching.

Prior to initiation of the experiment, pseudostem trapping was conducted to determine if *B. bassiana* was already present in the plots. Pseudostem trapping involved placing two split pseudostem pieces (30 cm long) at the base of each mat and checked after three days. A total of 120 banana weevil adults were collected from these traps. These were placed in Petri dishes (one per dish) with moist tissue paper, taken to the laboratory for incubation (21 days at 25–27°C and 80–90% r.h.) and checked for signs of infection every 3 days. Dead weevils observed in the field were also examined for mycosis (whitish fungal mycelial growth). Neither field-collected nor dead weevils showed signs of mycosis suggesting that *B. bassiana* was either not present or at negligible levels in the study fields prior to the onset of the trials.

Both inoculated and uninoculated weevils were released on all banana mats in the plots. Released weevils had been collected from the Masaka district, marked according to sex, mat of release, and whether or not they were to be inoculated with the fungus. Weevil sex was determined on the basis of punctuation on the rostrum (Longoria 1968) and curvature of the last abdominal segment (Roth and Willis 1963). Small scratch marks were placed on the head and elytra to denote sex and mat of release within the plots, respectively. Adult weevils were inoculated by placing them in 3-cm diameter Petri dishes containing 2 g of maize-formulated *B. bassiana* (approximately 3×10^9 conidia/g) for 6 h (Nankinga and Moore 2000). Exposed weevils were then placed in 9-cm Petri dishes lined with moist tissue paper and maintained in the laboratory (25–27°C, 80–90% r.h.) for 3 days to ensure infection before releasing them in the field. Previous studies using the same fungal isolate and inoculation method showed infection rates to adult weevils exceeding 97% (Nankinga 1999). Ten inoculated weevils (five males and five females) were released per mat one week after the release of 16 uninoculated weevils (eight males and eight females) per mat in the same plot. Releases were made during evening hours.

Pseudostem trapping was conducted 7 and 14 days after the release of the inoculated weevils in the two plots. Two pseudostem pieces were placed at the base of each mat and inspected after one day for *C. sordidus*. For marked weevils found in traps, we determined their mat of release and calculated the distance they had moved.

Experiment 2. Distribution of inoculated C. sordidus adults in field cages and banana stands

Caged trial. This experiment was conducted during the period from July to September 2003 to determine locations in the field where dead *C. sordidus* due to *B. bassiana* infection are found. *Cosmopolites sordidus* were exposed to *B. bassiana* in the laboratory, released into field cages at Sendusu, and then recovered by systematic searching of different niches within the cages. The cages (L × W × H: 120 × 90 × 60 cm) were prepared out of polythene sheets and corner and side stakes. Each cage was sunk into a 30-cm depression in the ground. Soil and grass mulch were placed inside the cage to make it level with the ground surface. The sides of the cages were 30 cm higher than the soil level to reduce weevil escape. Twelve banana sword suckers (i.e. plants less than 4 months old) with 30-cm shoots were planted in each cage. The plants were spaced at 30 cm in a 3 × 4 arrangement. There were six replicate cages.

Cosmopolites sordidus adults were placed in 3-cm Petri dishes containing 2 g of maize-formulated *B. bassiana* for 6 h. Exposed weevils were then maintained in 9-cm Petri dishes lined with moist tissue paper for 3 days after which they were marked and released into the cages (five males, five females per plant) during evening hours.

Systematic searching for dead weevils with mycosis was carried out 2 weeks later. First we removed and sifted through the mulch and soil surface below the mulch. Then we uprooted and dissected each plant. The numbers of dead and live weevils were recorded for the following locations: (i) mulch, (ii) soil <0.1 m from the mat, (iii) soil >0.1 m from the mat, (iv) leaf sheaths (v) pseudostem, (vi) corm surface, and (vii) inside the corm.

Field trial. Non-destructive searches were conducted for *C. sordidus* released in *Experiment 1* at 2, 4 and 6 weeks after release. This included (i) sifting through the leaf trash which was then left *in situ*, (ii) closely examining the outer surfaces of banana residues, (iii) soil near mat, and (iv) plant base in the leaf sheath. Both the inoculated and uninoculated weevils that were found were recorded according to the locations of recovery. Weevils that were recovered without mycosis were placed in moist Petri dishes and taken to the laboratory to observe any signs of infection and mycosis. The percentage of recaptured weevils showing mycosis was calculated for each field location.

Experiment 3. Horizontal transmission of B. bassiana under field conditions

An experiment to evaluate horizontal transmission of *B. bassiana* from infected to uninfected *C. sordidus* under field conditions was carried out concurrently with *Experiment 1* in banana plots at Sendusu. As noted above, *B. bassiana* did not appear to be present in the field prior to the release of inoculated and uninoculated weevils.

Pseudostem trapping was conducted 7, 14, 21, 35 and 42 days after release of inoculated weevils. Two split pseudostem pieces were placed per mat and inspected after 3 days. For every sampling occasion, weevils captured were placed in individual vials that were labeled according to whether the weevil had been released inoculated or uninoculated or was part of the existing field population that had not been released. The weevils were then taken to the laboratory, placed individually in Petri dishes identified to date of collection, and incubated as described in *Experiment 1*. For each sampling occasion, we monitored the number of dead weevils showing signs of mycosis from each plot every 3 days for a period of 12 days.

Experiment 4. Pheromone-baited delivery systems of B. bassiana

At Sendusu, we evaluated the rate of transmission of *B. bassiana* to *C. sordidus* using a pheromone-baited trap as an inoculation device. The treatments involved using the pheromone in combination with *B. bassiana*. The fungus was applied: (i) inside the trap, (ii) around the pheromone trap mat, and (iii) around the pheromone trap mat as well as at the base of four adjacent mats. The controls consisted of fungus applied at the central mat without a pheromone trap. Each treatment was placed in a plot consisting of five rows of five plants (cv Atwalira) spaced at 3 m. Treatments were replicated three times. Trial 1 was begun in July 2003, while trial 2 was conducted in the same plots in January 2004. Prior to initiation of each trial, weevils were collected from each plot as described in *Experiment 1* and incubated in the laboratory to determine if *B. bassiana* was present in the plots. Weevils did not show signs of mycosis suggesting that *B. bassiana* was either not present or at negligible levels in the study fields prior to the onset of the trials.

Crushed-maize formulated *B. bassiana* (200 g, approximately 3×10^9 conidia/g) was applied by spreading the formulation at the base of each mat or placing it inside the inoculation trap (pheromone trap). A thin (<3 cm) layer of grass mulch (*Panicum maximum*) was applied over the fungus to prolong its viability (Nankinga 1999). In both trials, weevils were collected from farmers' fields using pseudostem traps and then maintained in the laboratory for 1 week before release. Five males and five females, marked according to their sex and mat of release, were released at each of the mats. Weevils were released during evening hours.

A gallon trap with a ramp that allows easy entry and exit of the weevils via a ramp leading in and out of the trap was used (Tinzaara, 2005). The trap was made out of a 5-L plastic jerry can. A window was cut in each side of the jerry can and the flap folded down to make a walk-in ramp. The gallon traps were placed at the centre of plots 2 days after weevil release.

Pseudostem trapping was conducted after 14, 28 and 42 days. The recaptured weevils were inspected for pathogen infection, and the distance from the site of recapture to the fungus source or pheromone source was recorded. The weevils were then placed in Petri dishes (one per dish) that were labeled according to distance of recapture (0 m, less than 5 m, and more than 5 m from the inoculation trap). Within 2 h of collection, the weevils were transferred to the laboratory for incubation as in *Experiment 1*. The number of dead weevils showing signs of mycosis from each plot or treatment at different distances was recorded every three days for a total of 12 days.

Statistical analysis

The association between the number of *B. bassiana* inoculated and uninoculated weevils with their range of distance moved was analysed using a contingency table test of statgraphics plus version 7 (StatPoint, Inc., USA). The number of weevils recovered by pseudostem trapping from plots with different pathogen delivery systems that showed mycosis after incubation were analysed using a contingency table test on numbers (pooled for two trials), followed by Bonferroni correction, $\alpha = 0.05/6 = 0.008$, for multiple comparisons.

Results

Experiment 1. Dispersal of inoculated and uninoculated *C. sordidus* under field conditions

For both plots, a total of 56 inoculated weevils (7%) and 495 uninoculated weevils (37%) were recovered in pseudostem traps (Table I). In addition, another 113 inoculated weevils (14%) and 147 uninoculated weevils (11%) were recovered by visual searching in banana plots. Uninoculated weevils tended to move further than inoculated weevils ($\chi^2 = 27.8$, $df = 4$, $P = 0.001$) (Table II). More uninoculated weevils (28%) than inoculated weevils (13%) moved more than 6 m from the point of release.

Experiment 2. Distribution of inoculated *C. sordidus* adults in field cages and banana stands

Cage trial. Of the 720 *B. bassiana* inoculated-weevils released into the six cages, 590 (82%) were recovered dead and with mycosis, 5 (<1%) were recovered alive and without signs of infection, and 125 weevils could not be accounted for (Table III). The five live weevils showed signs of mycosis within 7 days during incubation in the laboratory. Seventy-one percent of the dead weevils showing mycosis were recovered from the corm surface, while 18% were found at the base of the banana mat (i.e. in the leaf sheath and soil near mat). Five percent of the dead weevils with mycosis were found in the soil in alleys or in the mulch. More males than females were recovered in the leaf sheath ($\chi^2 = 7.1$, $df = 1$, $P = 0.008$) and in the soil in alleys ($\chi^2 = 4.3$, $df = 1$, $P = 0.04$). Equal numbers of males and females were recovered in each of the other locations.

Field trial. We recovered nearly twice as many dead as live weevils by searching in plots (Table IV). All dead weevils showed signs of mycosis. Significantly more dead weevils were recovered in the leaf sheaths at the base of the plant than in soil near mat ($\chi^2 = 30.1$, $df = 1$, $P = 0.0001$), banana residues ($\chi^2 = 35.0$, $df = 1$, $P = 0.0001$) and

Table I. Percentage of *B. bassiana*-inoculated and uninoculated banana weevils recovered from pseudostem traps and by searching after release in plots of East African Highland bananas at Sendusu, Uganda.

(a) Pseudostem trapping followed by incubation in the laboratory			
Treatment	Weevils observed after incubation		
	Alive	Dead (with mycosis)	% infection ¹
Released inoculated	19	37	66
Released uninoculated	456	39	7
Unmarked (field weevils)	838	37	4
(b) Weevils recovered by searching in plots			
Treatment	Weevils recovered in field		
	Alive	Dead (with mycosis)	% infection ²
Released inoculated	6	107	95
Released uninoculated	133	14	10
Unmarked (field weevils)	186	11	6

¹Pooled infection of total weevils for two trials during five sampling occasions. A total of 840 infected and 1344 uninoculated weevils were released in the two trials. ²Pooled infection of total weevils for two trials during three searching occasions.

Table II. Percentage of *B. bassiana* inoculated and uninoculated banana weevils recovered by pseudostem traps at different distances from the point of release in East African Highland bananas.

Distance moved (m)	% of weevils recovered	
	Inoculated weevils (<i>n</i> = 189)	Uninoculated weevils (<i>n</i> = 474)
0–3	71	51
3.1–6	16	21
6.1–9	6	16
9.1–12	2	7
>12	5	5

The data of five sampling occasions were pooled before analysis; *n* = total number of weevils recovered. There was an association between the ranges of distances moved and the infection of the banana weevil ($\chi^2 = 27.8$, *df* = 4, *P* = 0.001, 5 × 2 contingency table test).

leaf trash ($\chi^2 = 46.2$, *df* = 1, *P* = 0.0001). Equal numbers of dead weevils were found in the soil near the base of the plant, in crop residues and leaf trash. Male and female weevils recovered with mycosis showed a similar distribution in different niches.

Experiment 3. Horizontal transmission of B. bassiana under field conditions

Fifty-six (7%) inoculated weevils and 495 (37%) uninoculated weevils were recovered in pseudostem traps. In addition, 457 weevils were recovered by searching of the plot: 113 (25%) were inoculated, 147 (32%) were uninoculated and 197 (43%) were unmarked field weevils (Table I). Of the weevils recovered by searching, 6–10% of the unmarked field weevils and of those that had been released uninoculated died due to pathogen infection. Four to seven percent of the uninoculated weevils that were recaptured by pseudostem trapping died due to pathogen infection after incubation in the laboratory for 12 days.

Experiment 4. Pheromone-baited delivery systems of B. bassiana

More weevils died due to *B. bassiana* infection in plots where *B. bassiana* was applied around the inoculation trap and four adjacent mats (13%) than where the pathogen was only applied inside or around the trap (5%) (Figure 1). This was true for all

Table III. Number of banana weevils with mycosis recovered in different locations 2 weeks after laboratory inoculation and release into cages at Sendusu, Uganda.

Location	Number of weevils dead with mycosis recovered in each location		
	Male	Female	Total
Corm surface	191	226	417
Soil near mat	26	28	54
Leaf sheath	35	16	51
Inside corm	20	14	34
Soil in alleys	14	5	19
Mulch	5	6	11
Pseudostem	2	2	4
Total	293	297	590

¹A total of 720 weevils were released in six cages. Five weevils were recovered alive.

Table IV. The numbers of banana weevils showing mycosis due to *B. bassiana* infection of the total weevils released inoculated and uninoculated that were recovered by hand searching in different locations in the banana field.

Location of recapture	Weevils ¹ recovered dead with mycosis			Weevils ¹ recovered alive		
	Total	Male	Female	Total	Male	Female
Plant base–leaf sheath	100	48	52	44	28	16
Soil near mat	36	19	17	21	12	9
Banana residues	32	16	16	25	15	10
Leaf trash	25	15	10	8	4	4
Total	193	98	95	98	59	39

¹Pooled number of weevils (inoculated and uninoculated) recovered in different locations during three searching occasions.

distances from the delivery site of the pathogen. There were more weevils showing signs of mycosis that were recovered more than 5 m from the pathogen delivery site when the pathogen was applied on the pheromone trap mat and on adjacent mats (39%) than when the pathogen was applied alone (10%). Equal numbers of dead weevils were recovered when the pathogen was applied inside compared to when it was applied around the pheromone trap mat. The number of weevils infected was significantly higher when the pathogen was applied inside the pheromone trap and

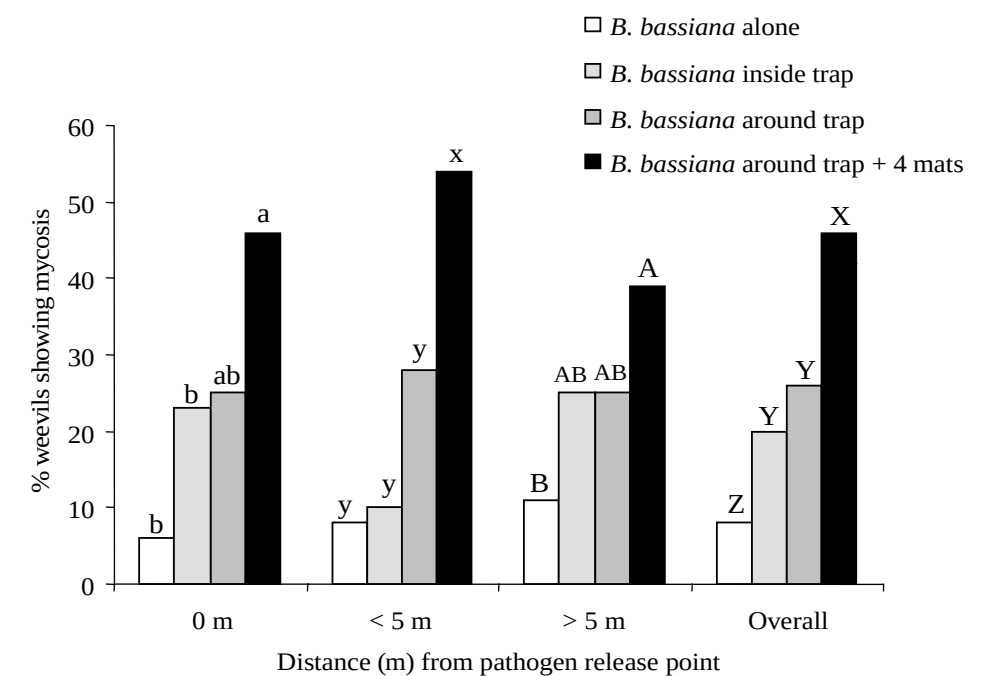


Figure 1. Percentage of banana weevils that died and showed signs of mycosis after incubation for 21 days and recaptured with pseudostem traps at different distances from the *Beauveria bassiana* release point (inoculation trap or central mat) in banana plots. The bars followed by similar letters are not significantly different (tested per distance from pathogen release point): 4×2 contingency table test followed by Bonferroni correction $\alpha = 0.05/6 = 0.008$. The total number of dead weevils is indicated in the bars.

around the pheromone trap mat than when the pathogen was applied without the pheromone trap.

Significantly more infected weevils were recovered by searching in plots where the pathogen was applied around the pheromone-baited trap plus four adjacent mats compared to all other treatments (Table V). In this treatment, there were more infected weevils (10) that were recovered >5 m from delivery site than when the pathogen was applied inside (2) and around the inoculation trap. No dead weevils were recovered beyond 5 m from the release point where pathogen was applied without the pheromone trap. The number of dead weevils showing mycosis recovered by searching was similar for plots where *B. bassiana* was applied inside versus around the inoculation trap.

Discussion

Entomopathogenic fungi such as *B. bassiana* have been proposed as potential control agents of banana weevil adults (Nankinga 1999; Godonou et al. 2000), but effective delivery of this pathogen into adult habitats is still a problem. Pheromone lures have been suggested as a means of aggregating banana weevils at delivery sites of fungal pathogens (Budenberg et al. 1993b; Tinzaara et al. 2002). Pheromone lures have been used successfully to spread entomopathogens to suppress populations of sweet potato weevils, *Cylas formicarius* (Coleoptera: Curculionidae) (Yasuda 1999) and bark beetles, *Ips typographus* (Coleoptera: Scolytidae) (Kreutz et al. 2004). In soil-borne insects such as banana weevil, attraction to pheromone lures would need to be coupled with horizontal transmission from infected to uninfected insects for adequate control. Our results indicate that an aggregation pheromone can be used to attract *C. sordidus* to sites where the entomopathogen *B. bassiana* is available for subsequent dissemination by the insect. In addition, we have demonstrated for the first time that horizontal transmission of *B. bassiana* does occur among *C. sordidus* under field conditions and can be enhanced by use of an aggregation pheromone.

Significantly more weevils were observed on the mat where the pheromone trap was placed than on adjacent mats, while the number of weevils at mats proximal to the pheromone trap mat was greater than on mats further away (Tinzaara et al. 2005). The aggregation effect is likely to be influenced by the efficacy of the pheromone trap. If the pheromone attracts a higher proportion of the population to the pathogen

Table V. Number of dead banana weevils due to *B. bassiana* infection that were recovered at different distances from the pheromone-baited inoculation trap or fungus release points in plots of East Africa Highland bananas.

Treatment	Number of weevils recovered by searching			
	0 m	<5 m	>5 m	Overall
<i>B. bassiana</i> alone	2	2	0	4c
<i>B. bassiana</i> inside trap	6	5	2	13b
<i>B. bassiana</i> around trap	7	4	2	13b
<i>B. bassiana</i> around trap + four mats	9	14	10	33a

The data presented in the table is pooled for two trials. Numbers of weevils in column “Overall” followed by similar letters are not significantly different: chi-square test followed by Bonferroni correction $\alpha = 0.05/6 = 0.008$.

delivery site, chances of transmission would be increased. Therefore, increasing pheromone efficacy will lead to higher transmission rates of *B. bassiana*. Theoretically, the strategy of using the pheromone to aggregate weevils on the pathogen delivery sites would have the advantage of reducing the amount of fungal pathogen applied per unit area compared to the technique of treating individual plants.

The effective control of the *C. sordidus* using the pheromone in a delivery system for *B. bassiana* assumes that infected individuals can disperse from the infection point and spread the pathogen throughout the field. In our study, up to 10% of uninoculated weevils that were released into an area where inoculated weevils had been earlier introduced became infected and died within 21 days after transfer to the laboratory. The result could have been an overestimation due to the incubation method and keeping weevils for a long time under laboratory conditions; infection may have been initiated post field collection. The percentage transmission seems to be low but it is possible that under suitable conditions (e.g., greater soil moisture, higher *C. sordidus* population levels) such a level of infection would increase the overall inoculum level. This fungus-carry-over-effect through the application of pheromone has been previously reported for sweet potato weevils as an effective way of spreading the fungal pathogens throughout a specific habitat (Yasuda 1999).

Most cadavers of infected weevils were found within the vicinity of the banana mat (i.e. on the corm, leaf sheath at the base of the mat and in the soil near the mat). These results have important implications for the use of this pathogen in the control of *C. sordidus*: after being infected with *B. bassiana* the weevils move to locations where healthy individuals are commonly found (Gold et al. 2004a). This increases the chances of pathogen transmission from infected to uninfected individuals. In laboratory studies Nankinga (1994) observed that transmission was greater from dead *C. sordidus* than from infected but still living ones.

An effective mechanism of transmission is a key factor in the ability of entomopathogens to develop epizootics (Roy and Pell 2000). The use of an attractant in the system as a method of introducing a deleterious agent into a pest population requires that the lured individuals can sufficiently disperse after visiting the self-contaminating site (Vega et al. 1995; Klein and Lacey 1999; Roy and Pell 2000). In our study using different delivery systems, it was observed that a number of weevils that died due to *B. bassiana* infection were recaptured more than 5 m from the pathogen source. This suggests that these weevils were contaminated with the pathogen from the delivery sites and dispersed after infection. Placement of the pathogen inside the inoculation device and on a few adjacent mats caused more infection to weevils compared to other delivery systems that were tested. The number of weevils dead per g of *B. bassiana* was however lower when the pathogen was placed inside the inoculation device and on a few adjacent mats than when the pathogen was applied inside and around the pheromone. The reason of the observed low infection or transmission of the fungal pathogen could have been due to the pheromone not being effective in attracting a large proportion of the weevils. It is also important to note that horizontal transmission might increase over time and that best laboratory strains might not be best performers in field.

Our study demonstrated that weevils aggregate around pheromone-baited traps. Application of *B. bassiana* in combination with pheromone-baited traps enhanced transmission of the pathogen from infected to uninfected individuals. Studies are currently in progress to evaluate the effect of integrating pheromones with *B. bassiana*

on weevil populations and damage. If the strategy is found promising, the data will be used in exploiting opportunities for integration of pheromones into a broader IPM programme to control the banana weevil. Subsequently, studies would be required to investigate the role of kairomones in enhancing transmission of fungal pathogens for the control of *C. sordidus*. Kairomones have the advantage that farmers can process them locally. Additional field-testing will be required to further validate the delivery system of *B. bassiana* using the aggregation pheromone and kairomones for the control of *C. sordidus*.

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